

[INTERNAL]

RAG System Architecture — Project Atlas

Technical Specification

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This document is the authoritative technical specification for the Retrieval-Augmented Generation system powering Project Atlas. It covers the end-to-end pipeline from query ingestion to cited response generation, the RBAC enforcement model, multi-collection routing, and the evaluation harness. The system went into private beta on 2026-03-01 and is on track for GA in Q3 2026 per the Product roadmap.

1. Pipeline Overview

The /ask endpoint accepts a query string and optional filters (department, classification). The pipeline: (1) validate RBAC — strip collections the user cannot access; (2) embed query with BAAI/bge-base-en-v1.5; (3) run hybrid search (dense + BM25) against permitted Qdrant collections; (4) fuse results with RRF k=60; (5) rerank top-20 with BAAI/bge-reranker-base; (6) pass top-5 chunks to Claude claude-sonnet-4-6 with a citation prompt; (7) stream response.

2. RBAC Enforcement

Users are assigned roles: **eng**, **product**, **exec**. Each role maps to permitted Qdrant collections and payload filter classifications. The mapping is stored in the AuthService and injected into the RequestContext at authentication time.

Role	Permitted Collections	Classifications
eng	eng_docs, stackoverflow	public, internal
product	prod_docs, eng_docs	public, internal
exec	all collections	public, internal, confidential

3. Citation Generation

The LLM prompt instructs the model to cite each factual claim with [Doc N] notation, where N corresponds to the source chunk index passed in the context. The API response includes a sources array with doc_id, title,

section, page, classification, and similarity_score for each cited chunk.

4. Evaluation Harness

The RAGAS evaluation suite runs against a 200-question golden dataset every Sunday at 03:00 UTC. Results are committed to the eval-results repository and surfaced on the Atlas Engineering dashboard. Current scores: faithfulness=0.89, answer_relevance=0.86, context_precision=0.82, context_recall=0.79. Target for GA: all metrics > 0.85.

- Faithfulness measures whether all claims in the answer are supported by the context.
- Answer relevance measures alignment of the answer with the original question.
- Context precision measures what fraction of retrieved chunks are actually relevant.
- Context recall measures coverage — are all relevant chunks being retrieved?